ICP Skill Factory — Approved Domain Map (Phase 1 Baseline)

Version: R5 (2026-03-13) **Status:** APPROVED for Phase 3 Entry **Change History:** See `phase1-review-changelog.md`, `phase1-formal-review-r3.md`, and `../02-skill-candidates/sources/R5-new-source-analysis.md`

本文件為整合 R3 審查修正及 R5 治理文件滾動更新後的正式基線。 Changes from the original Phase 1 map are marked with ★. R3 changes are marked with ◆. R5 governance enrichments are marked with ▲.

Level 1 — 14 Engineering Skill Domains (R3: D08/D10/D11/D14 修正)

#	Domain ID	Domain Name	Description
D01	OT-CYBERSECURITY	OT 資訊安全	OT/ICS security architecture, risk assessment, compliance, and protection
D02	SYSTEM-ARCHITECTURE	系統架構設計	OT/IT system architecture, network topology, interface design
D03	POWER-SYSTEM	電力系統工程	Power system analysis, design, simulation, renewable integration
D04	PROTECTION	保護工程	Protection coordination, relay engineering, fault analysis
D05	CONTROL-SYSTEM	控制系統工程	SCADA, EMS, DERMS, VPP design, configuration, tuning
D06	PANEL-ENGINEERING	盤櫃工程	Electrical panel design, wiring, terminal planning
D07	INTEGRATION	系統整合工程	Cross-system interface integration, protocol bridging, data exchange
D08	TESTING-COMMISSIONING	測試與試車	◆ FAT, SAT, commissioning, performance testing（D08.6 除役遷出至 D10.5）
D09	ENGINEERING-DOCS	工程文件管理	Technical documentation, design reports, document governance
D10	PROJECT-ENGINEERING	專案工程	★◆ **Post-acceptance** project technical management: requirements tracking, change management, technical coordination, contract technical management, system decommissioning
D11	ENGINEERING-GOVERNANCE	工程治理	Process standardization, quality control, design review, knowledge management

#	Domain ID	Domain Name	Description
D12	DATA-PLATFORM	能源資料平台	Energy data acquisition, storage, analytics, visualization
D13	ENGINEERING-AUTOMATION	工程自動化	Engineering automation tools, AI-assisted design, CI/CD
D14	PRE-GATE-ENGINEERING	前置技術工程	★ **NEW** Pre-Gate 0 → Gate 0 technical execution: requirement clarification, site survey, feasibility architecture, cost basis, Gate 0 input package

Level 2 — 73 Subdomains (R3: 修正 R2 算術錯誤 75→73)

D01 — OT 資訊安全 (7 subdomains, +1)

ID	Subdomain	Description
D01.1	安全架構設計	Zone/Conduit architecture, defense-in-depth, security zone partitioning
D01.2	風險評估與威脅建模	Asset identification, threat analysis, risk quantification. ▲ GOV-SD mandates triple-method integrated risk assessment: IEC 62443-3-2 + FMEA (RPN) + HAZOP. Risk source traceability required (T-XXX/FM-XXX/HAZ-XXX).
D01.3	合規與稽核	IEC 62443/ISO 27001 compliance assessment, gap analysis. ▲ GOV-SD defines SL Decision Lifecycle: Gate 0 propose → Gate 1 confirm → Gate 2 re-evaluate → Gate 3 validate.
D01.4	安全監控與事件回應	Real-time monitoring, anomaly detection, incident response
D01.5	安全加固與組態管理	Device hardening, access control, patch management
D01.6	供應鏈安全	Vendor risk assessment, component security verification
D01.7 ★	SIS 安全	Safety Instrumented System security isolation, SIS-specific controls

Boundary note for D01.7: Covers security controls specific to SIS/safety systems per IEC 62443 and IEC 61511 interface requirements. General OT security controls remain in D01.1–D01.6.

D02 — 系統架構設計 (6 subdomains, unchanged)

ID	Subdomain	Description
D02.1	OT 網路架構	Industrial network topology, VLAN planning, redundancy
D02.2	系統介面設計	Subsystem interface definition, data exchange specs
D02.3	通訊架構	Protocol selection and architecture-level communication design

ID	Subdomain	Description
D02.4	高可用與冗餘設計	Reliability design, fault tolerance, disaster recovery
D02.5	雲端與邊緣架構	Hybrid cloud, edge computing deployment
D02.6	架構評審與決策	ADR writing, technology selection evaluation

Boundary note for D02.3 (CHG-004): D02.3 covers architecture-level protocol selection and topology design. Device-level protocol configuration belongs to D05.5. Cross-system protocol translation belongs to D07.2.

D03 — 電力系統工程 (6 subdomains, unchanged)

ID	Subdomain	Description
D03.1	電力系統分析	Power flow, short circuit, stability analysis
D03.2	再生能源整合	Solar/wind grid integration, power forecasting
D03.3	儲能系統工程	BESS design, control strategy, economic assessment
D03.4	虛擬電廠	VPP architecture, DER aggregation, dispatch
D03.5	電力品質	Harmonics, power factor, voltage regulation
D03.6	電力系統模擬	System modeling, transient simulation, event replay

D04 — 保護工程 (4 subdomains, unchanged)

ID	Subdomain	Description
D04.1	保護協調	Protection coordination, relay settings
D04.2	繼電器工程	Relay selection, parameter setting, testing
D04.3	保護邏輯設計	Protection logic diagrams, interlock logic
D04.4	故障分析	Fault recording analysis, protection action assessment

D05 — 控制系統工程 (6 subdomains, unchanged)

ID	Subdomain	Description
D05.1	SCADA 系統設計	SCADA architecture, database, point list
D05.2	EMS / DERMS 配置	Energy/DER management system configuration
D05.3	HMI 設計	Human-machine interface design, alarm configuration
D05.4	PLC / RTU 程式設計	Controller programming and debugging
D05.5	通訊協定整合	Device-level protocol configuration, gateway setup
D05.6	控制策略設計	Automatic control logic, dispatch strategy, optimization

D06 — 盤櫃工程 (4 subdomains, unchanged)

ID	Subdomain	Description
D06.1	盤面佈局設計	Panel component layout, thermal planning
D06.2	配線設計	Wiring diagrams, terminal planning, wire labeling
D06.3	施工圖繪製	Fabrication drawings, cutout drawings, installation
D06.4	元件選型與規範	Component selection, technical specifications

D07 — 系統整合工程 (4 subdomains, unchanged)

ID	Subdomain	Description
D07.1	系統對接設計	Subsystem integration design and planning
D07.2	協定橋接與轉換	Cross-system protocol translation, gateway configuration
D07.3	資料整合	Cross-system data consolidation, normalization
D07.4	第三方系統整合	External system (ERP, GIS, weather) interfacing

D08 — 測試與試車 (5 subdomains) ◆ D08.6 遷出至 D10.5

ID	Subdomain	Description
D08.1	工廠驗收測試 (FAT)	Pre-shipment functional verification
D08.2	現場驗收測試 (SAT)	On-site functional and performance verification
D08.3	試車程序	System startup, step-by-step commissioning
D08.4	效能測試與驗證	Performance benchmarking, stress testing
D08.5	問題追蹤與缺陷管理	Defect tracking, severity classification, fix verification

FIX-001 說明：原 D08.6 系統除役（R5 生命週期活動）歸屬不當於測試域（R3 活動），已遷移至 D10.5。

D09 — 工程文件管理 (5 subdomains, unchanged)

ID	Subdomain	Description
D09.1	設計文件撰寫	Design reports, specifications, system descriptions
D09.2	圖面管理	Engineering drawings (SLD, wiring, layout) management
D09.3	會議與溝通記錄	Meeting minutes, action items, technical memos
D09.4	文件交付與版控	Document packaging, submission, version control
D09.5	技術寫作	Operation manuals, maintenance manuals, training materials

D10 — 專案工程 (5 subdomains) ★◆ RE-SCOPED to post-acceptance + decommissioning

Pre-gate subdomains (D10.1 需求分析, D10.2 技術可行性評估, D10.3 成本估算與報價, D10.6 售前技術支援) migrated to D14 per CHG-008. D08.6 系統除役遷入為 D10.5 per FIX-001。

ID	Subdomain	Description
D10.1 ★	專案需求管理	Post-contract requirements tracking, decomposition, traceability, scope baseline maintenance
D10.2 ★	變更管理	Design change evaluation, impact analysis, approval workflow
D10.3 ★	技術協調	Cross-department/vendor technical coordination, RFI management
D10.4 ★	合約技術管理	Technical scope tracking against contract, deliverable acceptance criteria, technical dispute resolution
D10.5 ◆	系統除役與資產處置	System decommissioning planning, data preservation, media sanitization, hardware disposal, R5 lifecycle management

Boundary rule (D10/D14): Gate 0 approval / contract award is the lifecycle boundary. D14 operates Pre-Gate 0 → Gate 0. D10 operates post-contract kickoff → project closure including decommissioning (R1–R5). D14.1 scope framing feeds D10.1 requirements baseline.

D11 — 工程治理 (6 subdomains, +1)

ID	Subdomain	Description
D11.1	設計審查	Design review, stage-gate review, approval. ▲ GOV-SD defines 4-gate control framework (Gate 0–3) with explicit blocking conditions per gate, 12-item Gate 3 delivery checklist.
D11.2	工程流程標準化	SOP definition, process documentation, continuous improvement. ▲ GOV-SDP establishes standards ownership system, exception ruling process, dispute escalation L1→L2→L3 with SLAs.
D11.3	品質管控	Quality planning, audit execution, nonconformance management. ▲ GOV-SDP defines 7 functional roles with RACI mapping across all gates, role-based KPI evidence model.
D11.4	知識管理	Knowledge capture, structuring, sharing, lessons learned
D11.5	標準與規範管理	Internal design standards, external standards tracking
D11.6 ★◆▲	工程能力管理	Engineering competency frameworks, training programs, qualification tracking（涵蓋安全能力作為首要實例化案例，並擴展至全領域工程能力認證；see ADR-006）. ▲ GOV-SDP provides SMART KPI definitions per role, RCW/AF/PVF scoring formula, annual governance cycle with data freeze and settlement.

D12 — 能源資料平台 (5 subdomains, unchanged)

ID	Subdomain	Description
D12.1	資料擷取與採集	Field device data acquisition, protocol parsing, edge processing

ID	Subdomain	Description
D12.2	資料儲存與管理	Time-series databases, data lakes, data lifecycle
D12.3	資料分析與建模	Energy data analytics, forecasting models, optimization
D12.4	視覺化與儀表板	Monitoring dashboards, reports, real-time visualization
D12.5	資料治理	Data standards, metadata management, access control

D13 — 工程自動化 (4 subdomains, unchanged)

ID	Subdomain	Description
D13.1	工程工具開發	Automated calculation, document generation, validation tools
D13.2	AI 輔助工程	AI/ML in engineering design, analysis, decision support
D13.3	CI/CD 與 DevOps	Automated build, test, deployment of engineering deliverables
D13.4	工作流程自動化	Repetitive workflow orchestration, automated reporting

D14 — 前置技術工程 (6 subdomains) ★ NEW

Pre-Gate 0 → Gate 0 technical execution. Primary role: Concept System Architect / Feasibility Owner. This is a technical execution domain, not a business development function.

ID	Subdomain	Description
D14.1 ◆	需求釐清與範圍框定	Customer/stakeholder requirement elicitation, scope boundary definition, ambiguity resolution, requirement conflict resolution, preliminary requirement traceability
D14.2 ◆	現場勘查與技術探勘	Physical site assessment, existing infrastructure inventory（含資產基線）, environmental constraints, brownfield integration assessment
D14.3 ◆▲	可行性評估與概念架構	Technical feasibility analysis, concept system architecture, alternative comparison analysis, technology selection, POC design, preliminary Zone/Conduit concept. ▲ GOV-SD defines Pre-Gate Design Support role producing 5 Gate 0 inputs; output is advisory (non-binding).
D14.4 ▲	成本基礎與 BOM 工程	Engineering-grade cost estimation, preliminary CBOM, labor hour baseline, vendor pricing, cost risk contingency. ▲ GOV-SD: CBOM = commercially quotable / design non-binding; EBOM supersedes at Gate 1. CBOM↔EBOM delta is business unit responsibility.
D14.5 ◆	利害關係人分析與介面清冊	Stakeholder identification and expectation alignment, cross-system interface point enumeration, legacy system capability assessment, integration constraint summary
D14.6	前置風險評估與 Gate 0 輸入包	Preliminary HLCRA, security classification input, risk-informed scope recommendations, Gate 0 decision package assembly

FIX-003 說明：原 D14.5「基線清冊準備」與 D14.2「現場勘查」存在資產盤點功能重疊，已重定位為「利害關係人分析與介面清冊」，補充原 D14 缺少的利害關係人管理面向。原資產基線功能併入 D14.2。

D14 dependency connections: D14→D01 (HLCRA/security classification feeds TRA), D14→D02 (concept architecture feeds detailed design), D14→D09 (preliminary docs enter governance pipeline), D14→D10 (scope/cost baseline feeds project execution), D14→D11 (Gate 0 package feeds gate review). See `phase1-revision-r2.md` for full dependency diagram.

Boundary Notes (R3 新增 ◆; R5 新增 ▲)

Boundary note for D03.4/D05.2 (REC-002): D03.4 涵蓋 VPP 的電力系統面（聚合策略、調度邏輯、市場參與規則），D05.2 涵蓋 VPP 的控制系統面（DERMS 軟體配置、通訊設定、即時控制迴路）。區分原則：標的為「電力系統行為」歸 D03.4，標的為「控制軟體配置」歸 D05.2。

Boundary note for D12.3/D13.2 (REC-001): D12.3 涵蓋以能源資料為標的的分析建模（含使用 AI/ML 作為工具），D13.2 涵蓋以工程流程為標的的 AI 輔助能力（如 AI 輔助設計審查、自動化文件生成）。區分原則：標的為「資料」歸 D12，標的為「工程流程」歸 D13。

▲ Boundary note for D01.2 (BOUNDARY-009, from GOV-SD): Design-time authority (Pre-Gate 0 through Gate 3) belongs to System Design Department. Execution-time (post-Gate 3 handover) belongs to project execution unit. Gate 3 handover = clean responsibility transfer; design approval ≠ risk acceptance (separate signers).

▲ Boundary note for D14.4 (BOUNDARY-010, from GOV-SD): CBOM (D14.4) = commercially quotable / design non-binding; produced by Pre-Gate Design Support. EBOM = design-binding; produced by System Architect at Gate 1+. CBOM↔EBOM delta is business unit responsibility, NOT design department responsibility.

Approved Map Statistics (R5)

Metric	Original	R1	R2	R3	R5
Level 1 Domains	13	13	14	14	**14**
Level 2 Subdomains	66	69	~~75~~ 73	73	**73**
Boundary clarifications	0	3	4	6	**10** (+BOUNDARY-007/008/009/010)
ADRs	3	4	5	6	**7** (+R5-ADR-001 Source Authority Hierarchy)
Source documents	3 + PRAC	—	—	—	**19 + PRAC + 2 GOV**
Source tiers	1	—	—	—	**3** (Tier 1 Governance / Tier 2 Exemplar / Tier 3 Contextual)

◆ERR 勘誤：R2 原宣稱 75 個子領域，實際加總為 73（D14 +6, D10 −2 = 淨 +4；69 + 4 = 73）。R2 誤將淨變動計為 +6，忽略 D10 縮減 −2。R3 修正。